

UNIT 5: MECHANISMS & PATTERNS OF EVOLUTION

5.4 – PRINCIPLES OF TAXONOMY & SYSTEMATICS

Biology 9

Mr. Queenan

Ch. 18.4 (pg. 596-599) & Ch. 19.1-19.2 (pg. 612-618)

Systemics Biology

- **Taxonomy** is the branch of biology concerned with identifying, naming, and classifying organisms.
 - A natural system of classification reflects the evolutionary history of organisms.
 - Naming and identifying organisms began with the Greeks and Romans.
 - Aristotle classified organisms into groups such as horses, birds, and oaks
 - In the Middle Ages, organisms were described using Latin names

Systemics Biology

- In the mid-eighteenth century, Carolus Linnaeus developed the system of **binomial nomenclature**
 - ▣ First word is the genus name
 - ▣ Second word is the specific epithet
 - Refers to one species (of potentially many) within its genus
 - ▣ A species is referred to by the full binomial name (Genus species)
 - ▣ Genus name can be used alone to refer to a group of related species



b. *Lilium canadense*

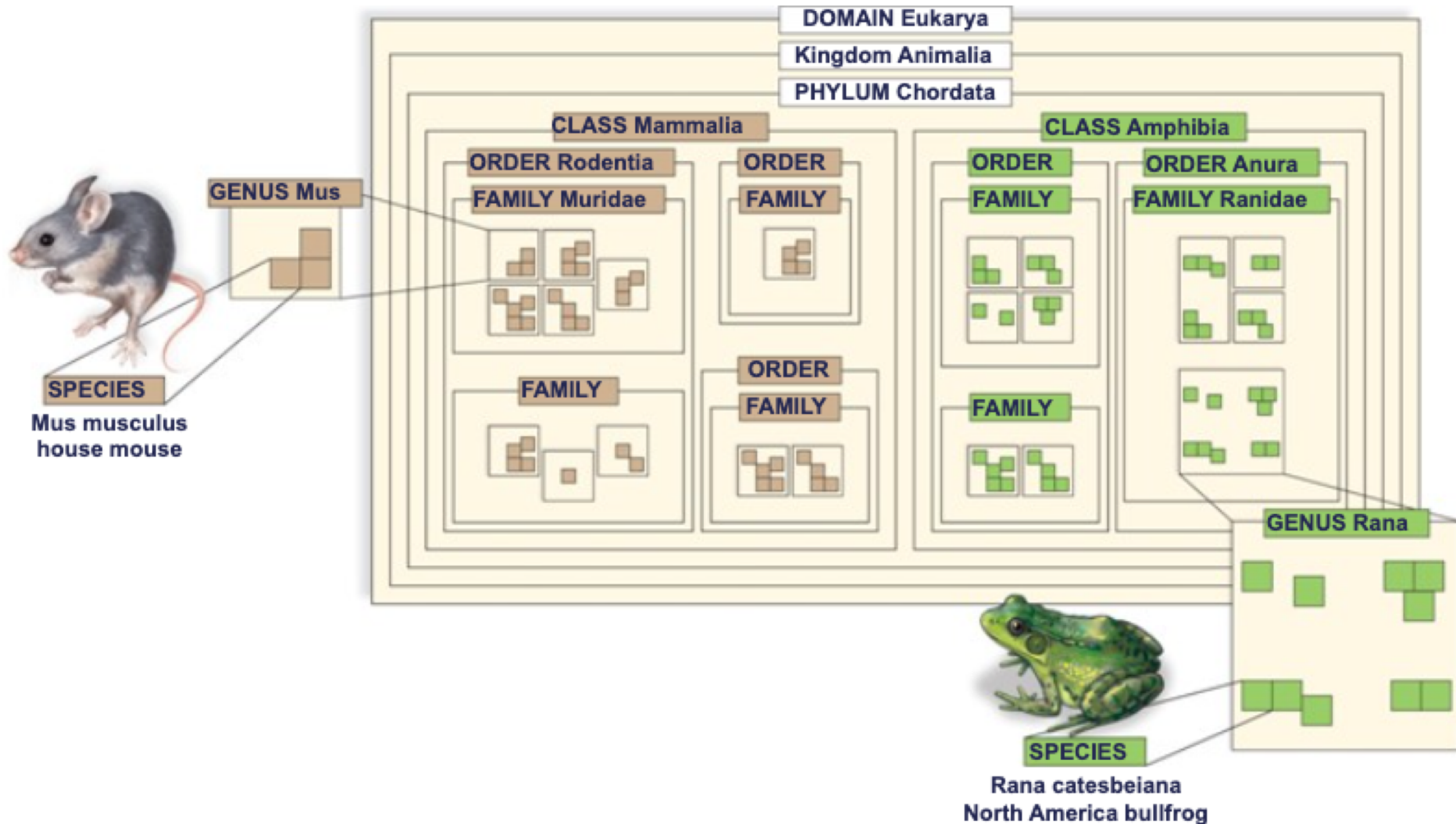


c. *Lilium bulbiferum*

Systemics Biology

- Who determines the scientific name? How are they derived?
 - ▣ Really up to the person who discovers a species that has not been previously named – Latin used because it was considered universal
 - *Alligator mississippiensis*
 - American Alligator, named for location
 - *Strigophilus garylarsonii*
 - Owl mite, named after cartoonist Gary Larson
- Modern taxonomists use the following classification:
 - ▣ **Species**
 - ▣ **Genus** – one or more species
 - ▣ **Family** – one or more genera
 - ▣ **Order** – one or more families
 - ▣ **Class** – one or more orders
 - ▣ **Phylum** – one or more classes
 - ▣ **Kingdom** – one or more phyla
 - ▣ **Domain** – one or more kingdoms

Systemics Biology



Systemics Biology

- The higher the category, the more inclusive
- Organisms in the same domain have general characteristics in common
- Members of a species share very specific characteristics.
- The task of creating standardized rules of nomenclature is difficult and has, most recently, been aided by the process of **DNA barcoding**
 - Compares short fragments of DNA sequences from an unknown organism to a large database of sequences from known organisms.
- Traditionally, taxonomists relied on anatomical data
 - Consortium for the Barcode of Life (CBOL), proposes that all scientists will be able to identify a species with the flick of a handheld scanner.
 - Like the 11-digit Universal Product Code (UPC) used in a supermarket, DNA is the UPC of organisms on Earth
 - A DNA—bar-coding device would provide a fast and inexpensive way to catalog organisms.

The Three Domain System

- Up until 20th century, only 2 kingdoms – *Plantae* or *Animalia*
 - ▣ By 1969, 5 kingdoms considered to encompass all life
 - ▣ Sequencing of rRNA suggests that all organisms evolved along three distinct lineages,
 - So different they needed to be categorized in a level broader than kingdom
- **Domain Bacteria**
 - ▣ Prokaryotic unicellular organisms that reproduce asexually.
 - ▣ Cyanobacteria are large photosynthetic prokaryotes.
 - ▣ Most bacteria are heterotrophic (break down organic remains)
 - ▣ Important in ecosystems - keeping chemical cycling going.
 - ▣ Some bacteria are parasitic and cause disease.

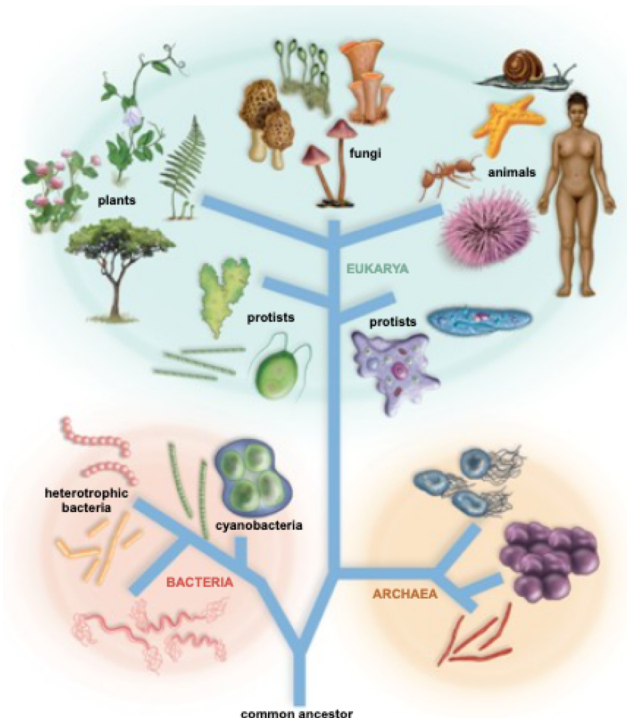
The Three Domain System

□ Domain Archaea

- Prokaryotic unicellular organisms that reproduce asexually.
- Live in extreme environments.
- Cell wall is diverse but not the same as the bacterial cell wall.

□ Domain Eukarya

- Unicellular and multicellular organisms
- Cells with a membrane-bounded nucleus
- Sexual reproduction is common
- Contains four kingdoms
 - Kingdom Protista
 - Kingdom Fungi
 - Kingdom Plantae
 - Kingdom Animalia

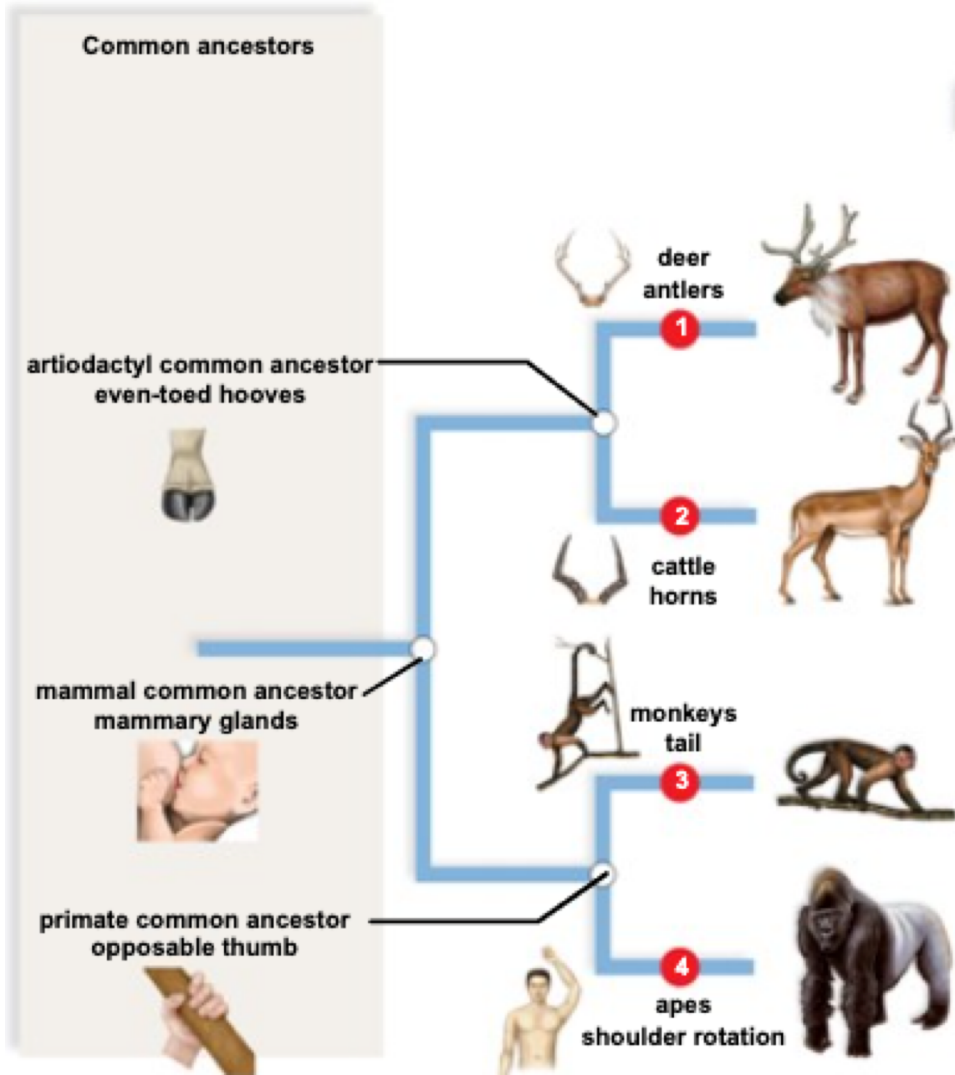


Phylogeny

- **Systematics** is the study of diversity of organisms using information from cellular to population levels
- One goal of systematics is to determine **phylogeny** (evolutionary history) of a group
- Phylogeny is often represented as a phylogenetic tree
 - ▣ A diagram indicating lines of descent
 - ▣ Each branching point:
 - Is a divergence from a **common ancestor**
 - Represents an organism that gives rise to two or more new groups
- Classification lists the unique characters of each taxon and is intended to reflect phylogeny
 - ▣ **Ancestral traits:**
 - Present in all members of a group, and
 - Present in the common ancestor
 - ▣ **Derived traits:**
 - Present in some members of a group, but absent in the common ancestor

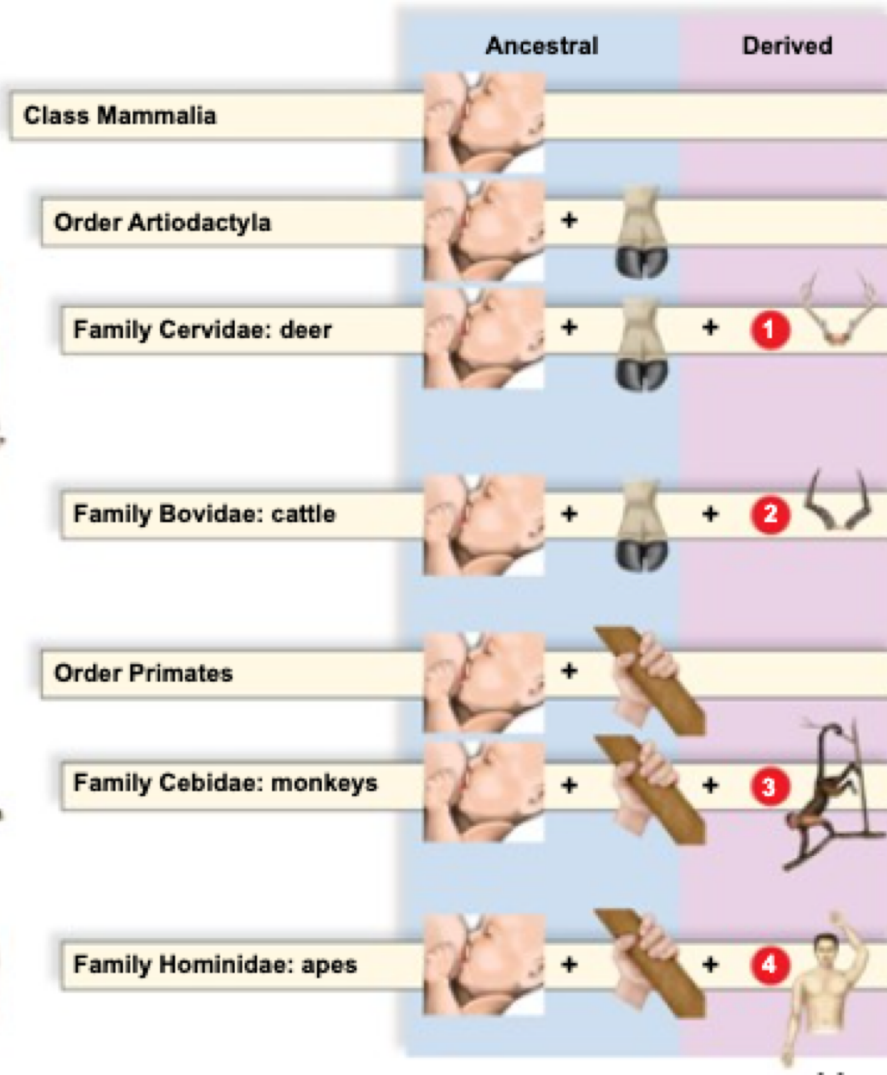
Phylogeny

Phylogeny



Classification

Trait Evolution

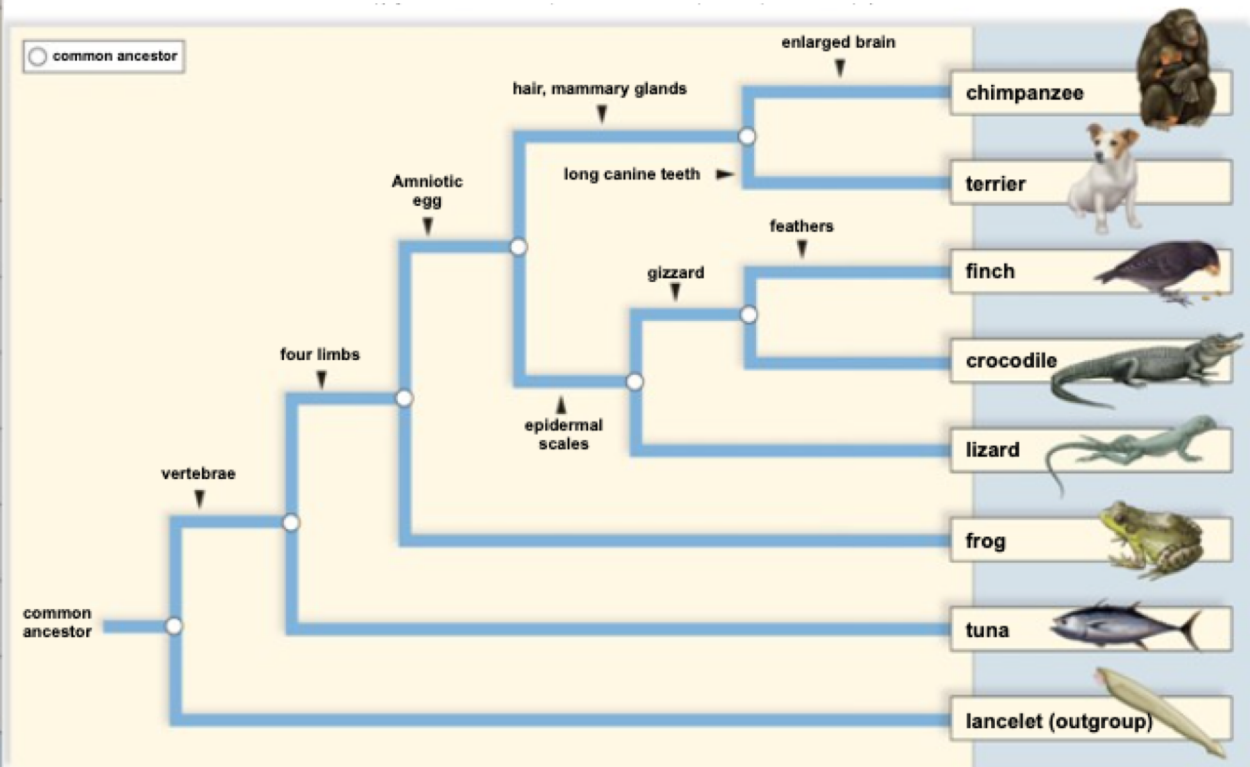


Phylogeny

- **Cladistics** is a way to analyze primitive and derived characters and by the construction of phylogenetic trees called a **cladogram** on the basis of shared derived characters.
 - ▣ Arrange taxa into a cladogram
- A cladogram is a special type of phylogenetic tree
 - ▣ A **clade** is an evolutionary branch that includes:
 - A common ancestor, together with all its descendent species
 - ▣ It traces the evolutionary history of the group being studied.
- Cladists are guided by the principle of **parsimony**—the minimum number of assumptions is most logical.
 - ▣ The best cladogram is one in which the fewest number of shared derived characters are left unexplained or that minimizes the number of assumed evolutionary changes.
- Reliability of cladograms is dependent on the knowledge and skill of an investigator

Phylogeny

Traits	Species							
	ingroup							lancelet (outgroup)
	chimpanzee	dog	finch	crocodile	lizard	frog	tuna	
mammary glands	X	X						
hair	X	X						
gizzard			X	X				
epidermal scales			X	X	X			
amniotic egg	X	X	X	X	X			
four limbs	X	X	X	X	X	X		
vertebrae	X	X	X	X	X	X	X	
notochord in embryo	X	X	X	X	X	X	X	X



Phylogeny

□ Tracing Phylogeny

▣ Fossil Traits

- Fossil record is incomplete
- It is often difficult to determine the phylogeny of a fossil

▣ Homology

- Refers to features that stem from a common ancestor
- **Homologous structures** are related to each other through common descent

▣ Analogy

- Similarity due to **convergent evolution**
- **Analogous structures** have the same function in different groups but do not have a common ancestry
 - Structures look similar due to adaptation to similar environments

Phylogeny

□ Tracing Phylogeny

▣ Behavioral Traits

- Parental care, mating calls, etc.

▣ Molecular Traits

- Systematics assumes:

- Two species with similar base-pair sequences are assumed to be closely related
- Two species with differing base-pair sequences are assumed to be only distantly related

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
Cow	c	c	c	c	g	t	g	g	a	g	g	t	a	c	g	c	t	t	c	a	c	t	c
Pig	c	c	c	c	g	t	g	g	a	g	g	t	g	c	g	c	t	t	c	a	c	t	c
Horse	t	c	c	g	g	t	g	g	a	g	g	t	g	c	g	c	t	t	c	g	c	c	c
Mouse	c	c	c	c	g	t	g	g	a	g	g	t	g	c	g	c	t	t	c	a	c	c	c
Rat	c	c	c	c	g	t	a	g	a	g	g	t	g	c	g	c	t	t	c	a	c	c	c
Dog	c	c	c	t	g	t	g	g	a	g	g	t	c	c	g	c	t	t	c	a	c	c	c
Guinea Pig	c	c	c	t	g	t	g	g	g	g	g	t	g	c	g	c	t	t	c	a	c	c	c
Chimp	c	c	t	g	g	t	g	g	g	g	c	t	a	c	g	c	t	t	c	a	c	c	t
Human	c	c	t	g	g	t	g	g	g	g	g	t	a	c	g	c	t	t	c	a	c	c	t
Orangutan	c	c	c	g	g	t	g	g	g	g	g	t	g	c	g	c	t	t	c	a	c	c	c
Macca	a	c	c	g	g	t	g	g	g	g	g	t	g	c	g	c	t	t	c	a	c	c	c

Phylogeny

□ Tracing Phylogeny

▣ Protein Comparisons

■ Immunological techniques

- Degree of cross reaction used to judge relationship

■ Amino acid sequencing

- Similar sequence in the same protein indicates a close relationship

▣ Molecular Clock

- Use neutral (non-adaptive) nucleotide sequences
- Assumes a constant rate of mutation over time

